

Self-replicating seedbox servers using programmable money

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Research Angle

Content distribution without a human operator

Centralized distribution is a single point of control

Mainstream content distribution concentrates control on one entity that owns both the hardware and the content.

- An operator decides what stays online. Availability depends on profitability, policies, and legality.
- Peer-to-peer protocols distribute the content across many nodes, but the hardware is still *managed* by humans.

The open research gap

Can the **infrastructure layer itself** be operator-independent, with no human controlling it after genesis?

Can a network of autonomous software agents, controlled only by inherited economic parameters, **survive** and **grow** without further human intervention?

Research question

The hypothesis under test: two outcomes must both hold, fleet **persistence** and **expansion**, with automated control and no human in the loop after genesis.

Falsifiable: the run makes one prediction, survive and grow, that the data can disprove.

Where this sits scientifically

Prior self-replicators ran on simulated resources:

- von Neumann's automata¹ and Ray's Tierra². Running on simulated CPU cycles and RAM.
- DAOs and autonomic computing self-manage, but only over virtual, quota-based resources.

Research Novelty: first self-replicating system on **real infrastructure paid for with real money**.



Three contributions:

- Engineering** — first self-replicating economic agent that buys real infrastructure on an open market.
- Method** — unmodified production code run against a replica Bitcoin network and a mock VPS marketplace.
- Empirical** — fleet dynamics and one heritable trait under selection.

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¹J. von Neumann (1966). *Theory of Self-Reproducing Automata*. Ed. by A. W. Burks. Urbana, IL: University of Illinois Press

²T. S. Ray (1991). "An approach to the synthesis of life". In: *Artificial Life II*. ed. by C. G. Langton, C. Taylor, J. D. Farmer, and S. Rasmussen. Redwood City, CA: Addison-Wesley, pp. 371–408

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The system

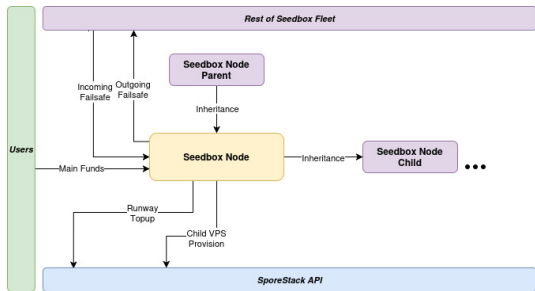
How a node lives, reproduces, mutates, and dies

System at a glance: one autonomous node

A node is one process running on a leased SporeStack (a Bitcoin-native marketplace) VPS with:

- **Bitcoin wallet** — pays its own bills.
- **Seedbox** — serves Creative Commons content.
- **IPv8 overlay** — gossips health telemetry to peers.
- **Decision logic** — the core actions.

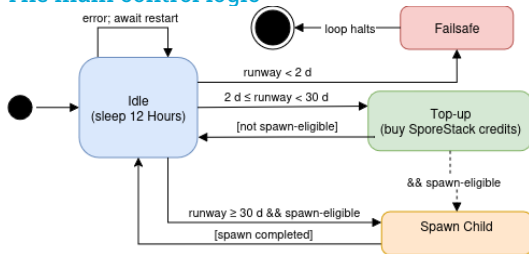
No central authority: bootstrapping genesis is the only human action. afterwards the fleet self-governs and no node has authority.



Arrows trace Bitcoin movements.

The decision loop

The main control logic



Every 12 h, the node takes the first action whose guard passes:

- Failsafe** — runway < 2 days → sweep remaining funds to the richest peer.
- Top-up** — runway < 30 days → buy VPS credits to cover another 30 days (*does not end tick*).
- Spawn** — if eligible → provision + fund a child.
- Idle** — wait for the next tick.

Identical code runs in every node.

Reproduction mechanics

Check post-spawn runway, not before

A node spawns only if its **post-spawn total runway** $\geq 60 \times (1 + \text{caution})$ days *and* its wallet can pay for the child's VPS. A node's **total runway** = VPS days + SporeStack credit + BTC.

- The child starts with a **30-day** pre-paid VPS and inherits **40%** of the Bitcoin remaining *after* the parent covers the child's expenses.
- Actual Pipeline: mutate caution $\rightarrow$ provision VPS $\rightarrow$ fresh BTC wallet $\rightarrow$ deploy code over SSH $\rightarrow$ transfer inheritance.

Caution — the single trait under selection.

Value $c \in [0.35, 0.9]$, with the genesis node starting at $c = 0.5$. Everything else is inherited.

The heritable trait

It scales the spawn threshold
 $60 \times (1 + c)$:

caution c	threshold
0.35	81 days
0.5	90 days
0.9	114 days

Low caution = spawn sooner
High caution = retain more

Mutation Formula:

$c =$

$c_{\text{parent}} + \mathcal{N}(0, 0.05) + 0.2(0.5 - c_{\text{parent}})$
Gaussian drift with reversion to 0.5.

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Methodology

Testing the node code offline and without cost

Faithful exercise of production code

Unchanged from live deployment

Node code is executed **identically** to a real deployment. Only changes relate to the compressed-time intervals: 1 sim-day $\approx$ 22 s.

Swapped the network layer only:

- Bitcoin $\rightarrow$ private **regtest**, node reuses same paths.
- SporeStack API $\rightarrow$ host-side **mock**, identical endpoints. Invoices paid in real regtest BTC.
- VPS $\rightarrow$ local **Alpine LXC container**. Parent deploys over the production SSH pipeline.
- Peer discovery via the same IPv8 tracker. An IPv8 bootstrap server keeps the fleet offline.

What runs locally is the code that would run on a live deployment.

Two experiments

No-income run — closed system

- Genesis seeded with **only €10,000**.
- Every node spends only its birth runway + inheritance.
- A **falsifiable closed-system prediction**: the fleet cannot sustain itself indefinitely.

Income run — open system

- Mock income pays the fleet
- Per-node income $\approx$ invariant (scales with node count).
- Uneven daily shares.
- **Node cap**: income suspended at 60 nodes, resumed at 40 $\rightarrow$ alternating growth/drought.

One caveat

Under load, the regtest drops $\sim$ half of all broadcasts, so income is set at $\sim 10\times$ **per-node rent**. **The income run tests fleet mechanics under favorable conditions, not its economic viability.**

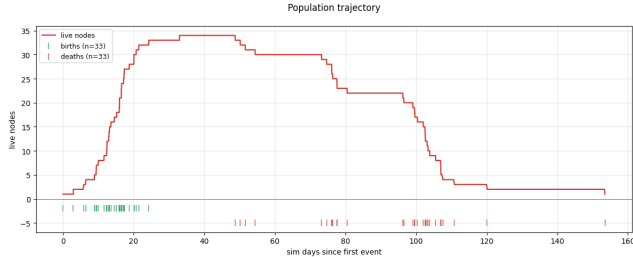
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Results

Mechanics confirmed, hypothesis tested

No-income run

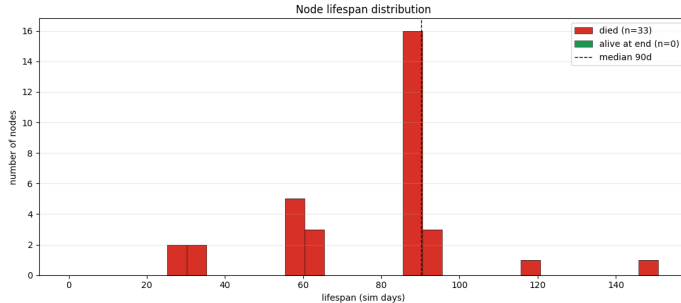
Fixed budget predictably spent?



- Every €3.27 buys exactly one day, so the fleet's **total lifespan** is at 3061 node-days (mean 93 days @ 33 nodes).
- One “old” survivor, as the failsafe concentrates residual funds into one node.

The closed-system prediction holds: a once-funded fleet cannot self-sustain.

The lifespan clusters were predictable



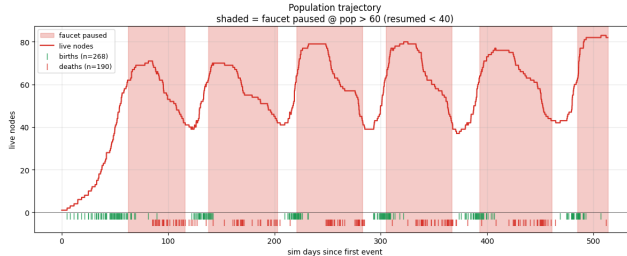
Each cluster is a spawn constant: **90 d** is the post-spawn runway any parent retains. **60 d** and **30 d** are the last two generations of children.

Specification met

The cluster positions should be fixed, with the histogram confirming.

Income run

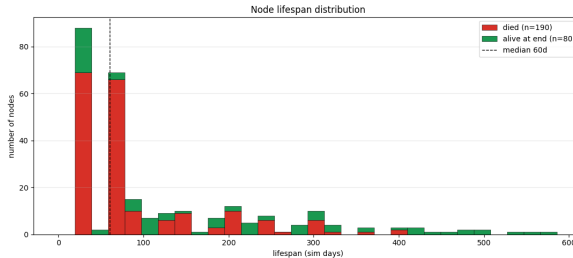
Open system



- The 40–80 swings are **caused by the node cap**, as income is cut at 60 nodes and restored at 40.
- Each spawn spreads capital over more nodes, shrinking per-node surplus. At these node counts, the cap masks it (→ future work).

Introducing income makes the fleet persist past 510 days.

Survival Distribution

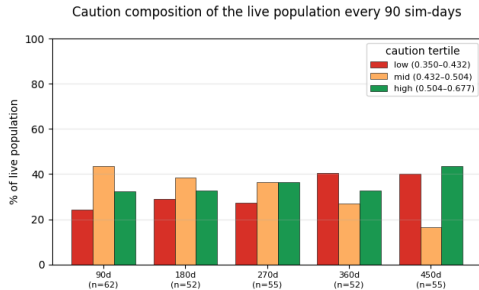


- Uneven early income splits the fleet, a node receiving more income early keeps large surplus, so **early wealth compounds** into a few long-lived nodes.

High turnover

Survival only needs *some* node holding a surplus, so a well-funded minority eventually replaces the short-lived majority.

Caution under selection: the hypothesis



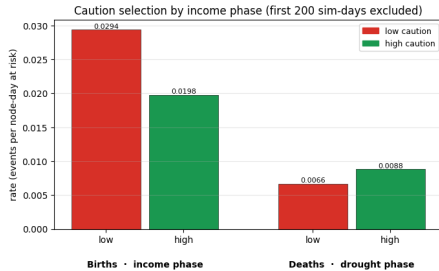
Caution should be **reverting toward 0.5**, yet mid-caution drops while both extremes grow. Means **selection pressure** is stronger than the artificial center-pull.

Hypothesis: the income on/off cycle rewards opposite strategies. Low caution exploits growth, high caution survives drought, while mid caution benefits from neither.

Two falsifiable predictions:

- a. low-caution reproduce faster during income.
- b. high-caution die less during pauses.

Caution under selection: the result



Prediction 1 – confirmed

Low-caution reproduces ~48% more with income (0.0294 vs. 0.0198 births/day).

Prediction 2 – disproved

High-caution has no survival advantage (0.0088 vs. 0.0066 deaths per day). Thus, selection favored low caution, the reproduction gap widened with every phase.

Income pauses were **too short** for caution to pay off. In hindsight, income should have been resumed with only 10-20 nodes alive. Testing on better host machine → future work.

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Viability & validity

Fleet economics, and the limits of the evidence

Not cost-competitive

Cost per TB-month

EternalSeedBox	€241
Commercial object storage	€20–25
Premium	≈ 10×

Break-even **€3.27 / node / day** (€98 per 30 d, 400 GB served).

Service becomes a deliberate niche

Each node pays for **retail VPS**, there is no economy of scale. The system buys availability that **survives takedown** and provides open-source accountability.

The market is **users whose content no one else will host**.

Competitive only when content availability is the product.

What the runs establish

Internal validity

- Unmodified production code against faithful **Bitcoin** and **SporeStack** replicas.
- Replication verified under **both** the closed and income conditions.
- The closed-system prediction was met.
- Genesis parameters are ultimately fully customizable.

External validity

- Forced node cap 60, not economics.
- Regtest drops $\sim$ transactions $\rightarrow$ income set $10\times$ rent $\rightarrow$ economy inflated and unmeasurable.
- Single VPS provider (SporeStack) – no fallback.
- Revenue mechanism not yet designed.
- User content mechanism not designed.

The mechanism is validated. Real-world viability is not.

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Conclusion

Limitations, and what comes next

Yes — conditional.

The fleet survives and grows after genesis **when its income covers or exceeds the per-node rent.**

The answer

Three contributions:

- a. **Engineering** — first self-replicating economic agent that buys real infrastructure on an open market.
- b. **Method** — unmodified production code against faithful Bitcoin + VPS-marketplace replicas.
- c. **Empirical** — fleet dynamics and one heritable trait under selection. A prediction met, one not observed.

Thank you!

Questions?

Future work

- **Payment incentive** — make consumers fund seedboxes directly.
- **Decentralized moderation** — solve removal requests across a fleet with no central authority.
- **Longer drought, no node cap** — to test the unconfirmed survival half of the caution hypothesis and better observe large-scale fleet dynamics.

Bibliography I



Neumann, J. von (1966). *Theory of Self-Reproducing Automata*. Ed. by A. W. Burks. Urbana, IL: University of Illinois Press.



Ray, T. S. (1991). "An approach to the synthesis of life". In: *Artificial Life II*. Ed. by C. G. Langton, C. Taylor, J. D. Farmer, and S. Rasmussen. Redwood City, CA: Addison-Wesley, pp. 371–408.